Semi-Supervised Triply Robust Inductive Transfer Learning (STRIFLE)

Tianxi Cai[a,b] Mengyan Li[c] Molei Liu[d]
Journal of the American Statistical Association (2025)

a,b Department of Biostatistics, Harvard University; c Department of Biomedical Informatics, Harvard University; d Department of Biostatistics, Columbia University

November 20, 2025

Content

1. Introduction

2. Background & Intuition

3. Derivation & Algorithms

4. Theoretical Results

5. Simulations & Real-Data Applications

Content

1. Introduction

2. Background & Intuition

3. Derivation & Algorithms

4. Theoretical Results

5. Simulations & Real-Data Applications

Real-Data: Mass General Brigham (MGB) Biobank

- Founded: 2008 (formally launched 2009)
- One of the largest hospital-based biobanks in the U.S., to ccelerate medical discovery through genomics and precision medicine.
- >150,000 participants enrolled.
- Contents.
  - **SNP**, genome-wide genotyping array data.
  - **EHR**, complete longitudinal electronic health records (diagnoses, medications, labs, imaging).
  - Biospecimens: Blood, DNA, serum, plasma.
  - Survey data: Environmental exposures, lifestyle factors
- **Access Policy**: **Not publicly available**, controlled access only.

Doubly Robust Estimators

Setup: Consider causal inference for an average treatment effect (ATE) with observed data (Y_i, A_i, X_i), where $\{A\}_{i\in[n]}$ is treatment, $\{X\}_{i\in[n]}$ are covariates, $\{Y\}_{i\in[n]}$ are outcomes.

Key idea: Combine two nuisance models

- Propensity score: $e(X) = P(A = 1 \mid X)$
- Outcome model: $m_a(X) = \mathbb{E}[Y \mid A = a, X]$

Estimator: Augmented Inverse Probability Weighted (AIPW)

$$\hat{\psi} = \frac{1}{n}\sum_{i=1}^{n}\left[\hat{m}_1(X_i) - \hat{m}_0(X_i) + \frac{A_i(Y_i - \hat{m}_1(X_i))}{\hat{e}(X_i)} - \frac{(1-A_i)(Y_i - \hat{m}_0(X_i))}{1-\hat{e}(X_i)}\right].$$

Property: Consistent if *either* $m_a(X)$ *or* $e(X)$ is correctly specified. (*doubly robust*)
Challenge: Bias $\propto \big(e(X) - \hat{e}(X)\big) \times \big(m(X) - \hat{m}(X)\big)$.

Surrogate-Assisted Semi-Supervised Learning (SAS)

Setup:

- Labeled data: (X_i, S_i, Y_i) for $i \in \mathcal{L}$ (small)
- Unlabeled data: (X_i, S_i) for $i \in \mathcal{U}$ (large)

S is a *surrogate variable* highly correlated with Y, e.g., lab tests or imaging signals in medical data. And usually $|\mathcal{L}| \ll |\mathcal{U}|$.

Goal: Use unlabeled (X, S) to improve estimation of $\mathbb{E}[Y \mid X]$.
Typical approach:

1. Fit imputation model $\hat{m}(X, S) \approx \mathbb{E}[Y \mid X, S]$
2. Plug in $\hat{m}(X, S)$ to fill missing labels and train the final predictor depending only on X

Benefit: Enhances efficiency when Y is scarce but surrogates S are abundant.
Challenge: Standard SAS assumes data comes from a single population. It cannot handle Covariate Shift or leverage external Source Data.

Transfer Learning

Setup: Two domains

$$\text{Source: } (X_S, Y_S) \sim \mathbb{P}_{\mathcal{S}}, \qquad \text{Target: } (X_T, Y_T) \sim \mathbb{P}_{\mathcal{T}},$$

where typically $\mathbb{P}_{\mathcal{S}} \neq \mathbb{P}_{\mathcal{T}}$.

Goal: Leverage source data to improve learning in the target domain (often with few labels).

Common strategies:

- **Importance weighting:** Reweight source samples by $w(x) = p_{\mathcal{T}}(x)/p_{\mathcal{S}}(x)$
- **Representation learning:** Learn domain-invariant features
- **Parameter transfer:** Use source model as initialization or regularization

Challenge: *Negative transfer*—source and target differ too much, leading to worse performance.

Content

1. Introduction

2. Background & Intuition

3. Derivation & Algorithms

4. Theoretical Results

5. Simulations & Real-Data Applications

Background & Intuition

Motivation from Biomedical Prediction:

- In genomics and medical studies, models are often trained on large well-labeled datasets (e.g., European ancestry populations) but need to generalize to under-represented groups such as **African American (AA)** cohorts, where labeled outcomes are scarce.
- This setting suffers from two main issues:
  1. **Covariate shift:** distributions of features (e.g., SNP or EHR variables) differ across groups.
  2. **Label scarcity:** outcome Y is costly or difficult to obtain in the target domain.

Core challenge: How to effectively leverage large labeled source data and abundant unlabeled target data to learn an unbiased and efficient model for the target population.

Intuition:

- Combine three ideas: **semi-supervised learning** (use unlabeled target data), **transfer learning** (borrow source information), and **double robustness** (remain consistent if either imputation model or weighting model is correct).
- The proposed **STRIFLE** framework achieves *triply robust* learning—robust to errors in imputation, weighting, or transfer modeling.

Content

1. Introduction

2. Background & Intuition

3. Derivation & Algorithms

4. Theoretical Results

5. Simulations & Real-Data Applications

Problem Setup

- $Y_i \in \mathbb{R}$, outcome variable, e.g., T2D status
- $\mathbf{X}_i \in \mathbb{R}^p$, candidate predictors including intercept, e.g., demographic variables and genetic markers (SNP).
- $\mathbf{S}_i \in \mathbb{R}^q$, surrogates for Y_i, e.g., EHR features like Hemoglobin A1C (HbA1c) test mentions
- $\mathbf{Z} = (\mathbf{X}^\mathsf{T}, \mathbf{S}^\mathsf{T})^\mathsf{T}$
- Index set L, where both Y and $\mathbf{Z}$ are observable. U, where only $\mathbf{Z}$ is observable.
- Index set $\mathcal{T}$ denotes target population, and $\mathcal{S}$ denotes source population.
- $p^{\mathcal{S}}_{Y|\mathbf{Z}}(y|\mathbf{z}) = p^{\mathcal{T}}_{Y|\mathbf{Z}}(y|\mathbf{z})$, but allow $p^{\mathcal{S}}_{\mathbf{Z}}(\mathbf{z}) \neq p^{\mathcal{T}}_{\mathbf{Z}}(\mathbf{z}) \Rightarrow p^{\mathcal{S}}_{Y|\mathbf{X}}(y|\mathbf{x}) \neq p^{\mathcal{S}}_{Y|\mathbf{X}}(y|\mathbf{x})$.
- Notation of sample size: $|\mathcal{I}_{\mathcal{S},L}| = n_{\mathcal{S}}$, $|\mathcal{I}_{\mathcal{T},L}| = n_{\mathcal{T}}$, $|\mathcal{I}_{\mathcal{S},U}| = N_{\mathcal{S}}$, $|\mathcal{I}_{\mathcal{T},U}| = N_{\mathcal{T}}$, $n = n_{\mathcal{T}} + n_{\mathcal{S}}$.
- Estimating goal: develop a prediction model of $Y|\mathbf{X}$ in $\mathcal{T}$ which satisfies

$$\mathbb{E}_{\mathcal{T}}\left[\mathbf{X}_i(Y_i - g(\mathbf{X}_i^\mathsf{T}\boldsymbol{\beta}))\right] = \mathbf{0}, \qquad \boldsymbol{\beta}_0 \in \mathbb{R}^p$$

Intuition

Overview

- To **leverage unlabeled data**, propose a working imputation model for $Y|\mathbf{Z}$.

$$m(\mathbf{Z}) = \mathbb{E}_{\mathcal{T}}(Y|\mathbf{Z}) = \mathbb{E}_{\mathcal{S}}(Y|\mathbf{Z}) = g(\mathbf{Z}^{\mathsf{T}}\boldsymbol{\theta})$$

For any fitted imputation model $\widehat{m}(\mathbf{Z}) = g(\mathbf{Z}^{\mathsf{T}}\widehat{\boldsymbol{\theta}})$, $\bar{\boldsymbol{\theta}} = \lim_{n\to\infty} \widehat{\boldsymbol{\theta}}$, $\bar{m}(\mathbf{Z}) = \lim_{n\to\infty} \widehat{m}(\mathbf{Z})$. Construct an SS estimator for $\boldsymbol{\beta}_0$,

$$\begin{aligned}\widehat{\boldsymbol{\beta}}(\widehat{\boldsymbol{\theta}}) &= \arg\min_{\boldsymbol{\beta}\in\Omega_{\boldsymbol{\beta}}} \mathcal{L}^{\boldsymbol{\beta}}_{N_{\mathcal{T}}+n_{\mathcal{T}}}(\boldsymbol{\beta};\widehat{\boldsymbol{\theta}}) + p_{\lambda_{\boldsymbol{\beta}}}(\boldsymbol{\beta}) \\ &= -\frac{1}{N_{\mathcal{T}}+n_{\mathcal{T}}}\Big[\sum_{i\in\mathcal{I}_{\mathcal{T},U}} Q(g(\mathbf{x}_i^{\mathsf{T}}\boldsymbol{\beta}), g(\mathbf{z}_i^{\mathsf{T}}\widehat{\boldsymbol{\theta}})) + \sum_{i\in\mathcal{I}_{\mathcal{T},L}} Q(g(\mathbf{x}_i^{\mathsf{T}}\boldsymbol{\beta}), y_i)\Big] + p_{\lambda_{\boldsymbol{\beta}}}(\boldsymbol{\beta})\end{aligned}$$

where Q is the quasi-log-likelihood function, i.e.,

$$Q\big(g(\mathbf{x}_i^{\mathsf{T}}\boldsymbol{\beta}), y_i\big) = y_i\mathbf{x}_i^{\mathsf{T}}\boldsymbol{\beta} - G(\mathbf{x}_i^{\mathsf{T}}\boldsymbol{\beta}) = y_i\mathbf{x}_i^{\mathsf{T}}\boldsymbol{\beta} - \int_0^{\mathbf{x}_i^{\mathsf{T}}\boldsymbol{\beta}} g(u')du'$$

- To **leverage source information**, propose a working density ratio model $w(\mathbf{z})$.
- To **avoid negative transfer**, include a transferability detection step.

Step 1: Estimating the Working Density Ratio Model

- Estimate $\frac{p_{\mathbf{Z}}^{(\mathcal{T})}(\mathbf{z})}{p_{\mathbf{Z}}^{(\mathcal{S})}(\mathbf{z})}$ using $w(\mathbf{z}) = \exp\{\mathbf{f}(\mathbf{z})^\top \boldsymbol{\zeta}\}$, $\mathbf{f} \in \mathbb{R}^L$ denotes for a vector of basis functions, thus can encapsulate a wide range of models.
- If $w(\mathbf{z})$ correctly specified, then $\mathbb{E}_{\mathcal{S}}\left[w(\mathbf{Z}_i)\mathbf{h}(\mathbf{Z}_i)\right] = \mathbb{E}_{\mathcal{T}}\left[\mathbf{h}(\mathbf{Z}_i)\right]$. (Lemma 1)
- Proof.

$$\mathbb{E}_{\mathcal{S}}[w_0(\mathbf{Z})h(Y,\mathbf{Z})] = \int\int h(y,\mathbf{z}) \cdot \frac{p_{\mathbf{Z}}^{(\mathcal{T})}(\mathbf{z})}{p_{\mathbf{Z}}^{(\mathcal{S})}(\mathbf{z})} \cdot p_{Y|\mathbf{Z}}^{(\mathcal{S})}(y|\mathbf{z}) \cdot p_{\mathbf{Z}}^{(\mathcal{S})}(\mathbf{z})\, dy\, d\mathbf{z}$$

$$= \int\int h(y,\mathbf{z}) \cdot p_{Y|\mathbf{Z}}^{(\mathcal{S})}(y|\mathbf{z}) \cdot p_{\mathbf{Z}}^{(\mathcal{T})}(\mathbf{z})\, dy\, d\mathbf{z}$$

$$= \int\int h(y,\mathbf{z}) \cdot p_{Y|\mathbf{Z}}^{(\mathcal{T})}(y|\mathbf{z}) \cdot p_{\mathbf{Z}}^{(\mathcal{T})}(\mathbf{z})\, dy\, d\mathbf{z} \quad \text{(by covariate shift assumption)}$$

$$= \mathbb{E}_{\mathcal{T}}[h(Y,\mathbf{Z})]$$

- Define $\boldsymbol{\zeta}_0 = \arg\min_{\boldsymbol{\zeta}\in\Omega_{\boldsymbol{\zeta}}} \left[\mathbb{E}_{\mathcal{S}}(\exp\{\mathbf{f}(\mathbf{z}_i)^\top\boldsymbol{\zeta}\}) - \mathbb{E}_{\mathcal{T}}(\mathbf{f}(\mathbf{z}_i)^\top\boldsymbol{\zeta})\right]$, empirically can be constructed through

$$\boldsymbol{\zeta}_0 = \arg\min_{\boldsymbol{\zeta}\in\Omega_{\boldsymbol{\zeta}}} \left[\frac{1}{N_{\mathcal{S}}} \sum_{i\in\mathcal{I}_{\mathcal{S},U}} \exp\{\mathbf{f}(\mathbf{z}_i)^\top\boldsymbol{\zeta}\} - \frac{1}{N_{\mathcal{T}}} \sum_{i\in\mathcal{I}_{\mathcal{T},U}} \mathbf{f}(\mathbf{z}_i)^\top\boldsymbol{\zeta}\right] + p_{\lambda_{\boldsymbol{\zeta}}}(\boldsymbol{\zeta})$$

Step 2-1: Construct a Target-Only Estimator of the Imputation Model

- Recall $\bar{\boldsymbol{\theta}} = \lim_{n\to\infty} \widehat{\boldsymbol{\theta}} := \arg\max_{\boldsymbol{\theta}} \mathbb{E}_{\mathcal{T}}\left[Q\{g(\mathbf{Z}_i^{\mathsf{T}}\boldsymbol{\theta}), Y_i\}\right] = \arg\max_{\boldsymbol{\theta}} \mathbb{E}_{\mathcal{T}}\left[y_i\mathbf{z}_i^{\mathsf{T}}\boldsymbol{\theta} - G(\mathbf{z}_i^{\mathsf{T}}\boldsymbol{\theta})\right]$.
- With $\mathcal{D}_{\mathcal{T},L}$ only, a straightforward way for constructing estimator of $\bar{\boldsymbol{\theta}}$ is

$$\widehat{\boldsymbol{\theta}}^{(\mathcal{T})} = \arg\min_{\boldsymbol{\theta}\in\Omega_{\boldsymbol{\theta}}^{(\mathcal{T})}} \Big[-\frac{1}{n_{\mathcal{T}}} \sum_{i\in\mathcal{I}_{\mathcal{T},L}} Q\{g(\mathbf{z}_i^{\mathsf{T}}\boldsymbol{\theta}), y_i\} + p_\lambda(\boldsymbol{\theta}) \Big]$$

- Under standard regularity conditions in high-dimensional settings, $\widehat{\boldsymbol{\theta}}^{(\mathcal{T})} \to \bar{\boldsymbol{\theta}}$.

Step 2-2: Construct a Meta-Learning Estimator of the Imputation Model

- Use $\mathcal{D}_{\mathcal{T},L} \cup \mathcal{D}_{\mathcal{S},L}$, we further define
$\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}\in\Omega_{\boldsymbol{\theta}}} \left\{ \frac{n_{\mathcal{S}}}{n}\mathbb{E}_{\mathcal{S}}\left[w_0(\mathbf{Z}_i)Q\{g(\mathbf{Z}_i^{\mathsf{T}}\boldsymbol{\theta}), Y_i\}\right] + \frac{n_{\mathcal{T}}}{n}\mathbb{E}_{\mathcal{T}}\left[Q\{g(\mathbf{Z}_i^{\mathsf{T}}\boldsymbol{\theta}), Y_i\}\right] \right\}$
- If $m(\mathbf{z})$ or $w(\mathbf{z})$ correctly specified, $\boldsymbol{\theta}^* = \bar{\boldsymbol{\theta}}$. Why?
- **If the density ratio model is correcty specified**, apply Lemma 1 with $h(Y, \mathbf{Z}) = Q\{g(\mathbf{Z}^{\mathsf{T}}\boldsymbol{\theta}), Y\}$,

$$\mathbb{E}_{\mathcal{S}}[w_0(\mathbf{Z}_i)Q\{g(\mathbf{Z}_i^{\mathsf{T}}\boldsymbol{\theta}), Y_i\}] = \mathbb{E}_{\mathcal{T}}[Q\{g(\mathbf{Z}_i^{\mathsf{T}}\boldsymbol{\theta}), Y_i\}]$$

Therefore

$$\begin{aligned}\boldsymbol{\theta}^* &= \arg\max_{\boldsymbol{\theta}} \left[\frac{n_{\mathcal{S}}}{n}\mathbb{E}_{\mathcal{T}}[Q\{g(\mathbf{Z}_i^{\mathsf{T}}\boldsymbol{\theta}), Y_i\}] + \frac{n_{\mathcal{T}}}{n}\mathbb{E}_{\mathcal{T}}[Q\{g(\mathbf{Z}_i^{\mathsf{T}}\boldsymbol{\theta}), Y_i\}]\right] \\ &= \arg\max_{\boldsymbol{\theta}} \left[\left(\frac{n_{\mathcal{S}} + n_{\mathcal{T}}}{n}\right)\mathbb{E}_{\mathcal{T}}[Q\{g(\mathbf{Z}_i^{\mathsf{T}}\boldsymbol{\theta}), Y_i\}]\right] = \arg\max_{\boldsymbol{\theta}} \mathbb{E}_{\mathcal{T}}[Q\{g(\mathbf{Z}_i^{\mathsf{T}}\boldsymbol{\theta}), Y_i\}] = \bar{\boldsymbol{\theta}}\end{aligned}$$

Step 2-2: Construct a Meta-Learning Estimator of the Imputation Model

- **If imputation model is correctly specified**, the moment condition holds:

$$\mathbb{E}_{\mathcal{T}}[\mathbf{Z}_i\{Y_i - \bar{m}(\mathbf{Z}_i)\}] = \mathbf{0}$$

which is equivalent to:

$$\mathbb{E}_{\mathcal{T}}[\mathbf{Z}_i Y_i] = \mathbb{E}_{\mathcal{T}}[\mathbf{Z}_i \bar{m}(\mathbf{Z}_i)]$$

The first-order condition for $\boldsymbol{\theta}^*$ is:

$$\frac{\partial}{\partial \boldsymbol{\theta}}\left[\frac{n_{\mathcal{S}}}{n}\mathbb{E}_{\mathcal{S}}[w_0(\mathbf{Z}_i)Q\{g(\mathbf{Z}_i^\top\boldsymbol{\theta}), Y_i\}] + \frac{n_{\mathcal{T}}}{n}\mathbb{E}_{\mathcal{T}}[Q\{g(\mathbf{Z}_i^\top\boldsymbol{\theta}), Y_i\}]\right]\Bigg|_{\boldsymbol{\theta}=\boldsymbol{\theta}^*} = \mathbf{0}$$

which simplifies to

$$\frac{n_{\mathcal{S}}}{n}\mathbb{E}_{\mathcal{S}}[w_0(\mathbf{Z}_i)\mathbf{Z}_i\{Y_i - g(\mathbf{Z}_i^\top\boldsymbol{\theta}^*)\}] + \frac{n_{\mathcal{T}}}{n}\mathbb{E}_{\mathcal{T}}[\mathbf{Z}_i\{Y_i - g(\mathbf{Z}_i^\top\boldsymbol{\theta}^*)\}] = \mathbf{0}$$

$$\mathbb{E}_{\mathcal{S}}[w_0(\mathbf{Z}_i)\mathbf{Z}_i\{Y_i - g(\mathbf{Z}_i^\top\boldsymbol{\theta}^*)\}] = \mathbb{E}_{\mathcal{T}}[\mathbf{Z}_i\{Y_i - g(\mathbf{Z}_i^\top\boldsymbol{\theta}^*)\}]$$

Therefore, the first-order condition becomes:

$$\mathbb{E}_{\mathcal{T}}[\mathbf{Z}_i\{Y_i - g(\mathbf{Z}_i^\top\boldsymbol{\theta}^*)\}] = \mathbf{0}$$

Since $m(\mathbf{z})$ is correctly specified, $\bar{\boldsymbol{\theta}}$ also satisfies:

$$\mathbb{E}_{\mathcal{T}}[\mathbf{Z}_i\{Y_i - g(\mathbf{Z}_i^\top\bar{\boldsymbol{\theta}})\}] = \mathbf{0}$$

Therefore, $\boldsymbol{\theta}^* = \bar{\boldsymbol{\theta}}$.

Step 2-2: Construct a Meta-Learning Estimator of the Imputation Model

- But if both models misspecified or $p_{Y|\mathbf{Z}}$ differs between $\mathcal{D}_{\mathcal{T}}$ and $\mathcal{D}_{\mathcal{S}}$, $\boldsymbol{\theta}^* \neq \bar{\boldsymbol{\theta}}$.
- Intuition: use a bias correction term $\boldsymbol{\delta}$ (residual) to fix the bias.

Step 2-2: Construct a Meta-Learning Estimator of the Imputation Model

Algorithm 1: Robust Meta-Learning of $\bar{\boldsymbol{\theta}}$.

- **Step 1.** Using $\mathcal{D}_{\mathcal{S},L} \cup \mathcal{D}_{\mathcal{T},L}$ along with $\widehat{w}(\mathbf{z}) = \exp(\mathbf{z}^\mathsf{T}\widehat{\boldsymbol{\zeta}})$, we estimate $\boldsymbol{\theta}^*$ as $\tilde{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}\in\Omega_{\boldsymbol{\theta}}} \mathcal{L}_n(\boldsymbol{\theta};\widehat{\boldsymbol{\zeta}}) + p_{\lambda_{\boldsymbol{\theta}}}(\boldsymbol{\theta})$, where

$$\mathcal{L}_n(\boldsymbol{\theta};\boldsymbol{\zeta}) = -\frac{1}{n}\sum_{i\in\mathcal{I}_{\mathcal{S},L}\cup\mathcal{I}_{\mathcal{T},L}} w(\mathbf{z}_i;\boldsymbol{\zeta})^{I(i\in\mathcal{I}_{\mathcal{S},L})} Q\{g(\boldsymbol{\theta}^\mathsf{T}\mathbf{z}_i), y_i\},$$

  $I(i \in \mathcal{I}_{\mathcal{S},L})$ is an indicator function, $p_{\lambda_{\boldsymbol{\theta}}}(\boldsymbol{\theta})$ is a penalty function on $\boldsymbol{\theta}$ with a tuning parameter $\lambda_{\boldsymbol{\theta}}$, and $\Omega_{\boldsymbol{\theta}}$ is the feasible parameter space.

- **Step 2.** Estimate $\boldsymbol{\delta}_0$ using $\mathcal{D}_{\mathcal{T},L}$ with $\tilde{\boldsymbol{\theta}}$ as $\widehat{\boldsymbol{\delta}} = \arg\min_{\boldsymbol{\delta}\in\Omega_{\boldsymbol{\delta}}} \mathcal{L}^{(\mathcal{T})}_{n_{\mathcal{T}}}(\tilde{\boldsymbol{\theta}} + \boldsymbol{\delta}) + p_{\lambda_{\boldsymbol{\delta}}}(\boldsymbol{\delta})$, to construct the meta-learning estimator $\widehat{\boldsymbol{\theta}}^{(\text{meta})} = \tilde{\boldsymbol{\theta}} + \widehat{\boldsymbol{\delta}}$, where

$$\mathcal{L}^{(\mathcal{T})}_{n_{\mathcal{T}}}(\tilde{\boldsymbol{\theta}} + \boldsymbol{\delta}) := -\frac{1}{n_{\mathcal{T}}}\sum_{i\in\mathcal{I}_{\mathcal{T},L}} Q[g\{\mathbf{z}_i^\mathsf{T}(\tilde{\boldsymbol{\theta}} + \boldsymbol{\delta})\}, y_i],$$

  $p_{\lambda_{\boldsymbol{\delta}}}(\boldsymbol{\delta})$ is a penalty function on $\boldsymbol{\delta}$ with a tuning parameter $\lambda_{\boldsymbol{\delta}}$, and $\Omega_{\boldsymbol{\delta}}$ is the feasible parameter space.

Step 3: STRIFLE Estimator of the Imputation Model

Motivation: Overcoming Negative Transfer

- **Problem with Meta-learning estimator $\widehat{\boldsymbol{\theta}}^{(\mathbf{meta})}$:** The second bias correction step (Algorithm 1, Step 2) is unable to fully correct the bias when $\boldsymbol{\delta}_0$ is too dense to be fully recovered with limited labeled target data, which leads to **negative transfer**: $\widehat{\boldsymbol{\theta}}^{(\text{meta})}$ performs worse than target-only estimator $\widehat{\boldsymbol{\theta}}^{(\mathcal{T})}$.
- **Solution: Adaptive transferability detection**
  Define STRIFLE estimator as a **convex combination**:

$$\widehat{\boldsymbol{\theta}}^{(\text{STRIFLE})} = \widehat{\rho}\widehat{\boldsymbol{\theta}}^{(\text{meta})} + (1-\widehat{\rho})\widehat{\boldsymbol{\theta}}^{(\mathcal{T})}$$

  where $\widehat{\rho} \in \{0, 1\}$ is a binary coefficient estimated via **data splitting**.

Step 3: STRIFLE Estimator of the Imputation Model

Step 1: Data Splitting

- Randomly split target labeled data $\mathcal{I}_{\mathcal{T},L}$ into two halves: $\mathcal{I}_{\mathcal{T},L}^{(1)}$ and $\mathcal{I}_{\mathcal{T},L}^{(2)}$.

Step 2: Construct Estimators on Each Fold

For $k = 1, 2$:

- Use data in $\mathcal{I}_{\mathcal{T},L}^{(k)}$ to construct $\widehat{\boldsymbol{\theta}}^{(\mathcal{T},k)}$ (target-only)
- Use data in $\mathcal{I}_{\mathcal{S},L}$ and $\mathcal{I}_{\mathcal{T},L}^{(k)}$ to construct $\widehat{\boldsymbol{\theta}}^{(\text{meta},k)}$ (via Algorithm 1)

Step 3: Compare Loss on Hold-out Data

For each fold k, evaluate loss on the *remaining* data $\mathcal{I}_{\mathcal{T},L} \backslash \mathcal{I}_{\mathcal{T},L}^{(k)}$:

$$\mathcal{L}_{n_{\mathcal{T}}/2}^{\rho(-k)}(\boldsymbol{\theta}) = \frac{2}{n_{\mathcal{T}}} \sum_{i \in \mathcal{I}_{\mathcal{T},L} \backslash \mathcal{I}_{\mathcal{T},L}^{(k)}} G(\mathbf{z}_i^{\mathsf{T}} \boldsymbol{\theta}) - y_i \mathbf{z}_i^{\mathsf{T}} \boldsymbol{\theta}$$

Step 4: Cross-Fitted Decision (Equation 9)

$$\widehat{\rho} = I\left\{ \frac{1}{2} \sum_{k=1}^{2} \mathcal{L}_{n_{\mathcal{T}}/2}^{\rho(-k)}(\widehat{\boldsymbol{\theta}}^{(\text{meta},k)}) \leq \frac{1}{2} \sum_{k=1}^{2} \mathcal{L}_{n_{\mathcal{T}}/2}^{\rho(-k)}(\widehat{\boldsymbol{\theta}}^{(\mathcal{T},k)}) - \epsilon_0 \right\}$$

where $\epsilon_0 \geq 0$ is a nonnegative constant chosen to achieve theoretical guarantee.

Tribly-Robustness

- First robustness (Classical Double Robustness). If either the imputation model or the density ratio model is correctly specified, we have $\delta_0 = 0$. Relating to Theorem 1, we further derive
$$\|\hat{\theta}^{\text{meta}} - \bar{\theta}\|_1 = O_p(\sqrt{s_\theta n^{-1} \log d})$$
- Second robustness: If both nuisance models are wrong but not "so wrong", i.e., $\|\delta_0\|_1 \lesssim \sqrt{s_\theta n_{\mathcal{T}} / \log d}$, the residual correction term can correct the bias, giving us
$$\|\hat{\theta}^{\text{meta}} - \bar{\theta}\|_2^2 \approx s_\theta n^{-1} \log d + \sqrt{n_{\mathcal{T}}^{-1} \log d} \|\delta_0\|_1 \ll s_\theta n_{\mathcal{T}}^{-1} \log d$$
- Third robustness, even if $\|\delta_0\|_1 \gg \sqrt{s_\theta n_{\mathcal{T}} / \log d}$, STRIFLE can help us
$$\|\hat{\theta}^{\text{STRIFLE}} - \bar{\theta}\|_2^2 \leq Err_{\text{Meta}} \wedge \{s_\theta n_{\mathcal{T}}^{-1} \log d + n_{\mathcal{T}}^{-1/2}\}$$

Content

1. Introduction

2. Background & Intuition

3. Derivation & Algorithms

4. Theoretical Results

5. Simulations & Real-Data Applications

Assumptions

Assmption 1: Convergence Rate of Density Ratio

Denote by Ω_0 the event that $\|\widehat{\boldsymbol{\zeta}} - \boldsymbol{\zeta}_0\|_1 \leq C_\zeta$, and $\|\widehat{\boldsymbol{\zeta}} - \boldsymbol{\zeta}_0\|_2^2 = o(1)$, where C_ζ is a positive constant, and $\Pr(\Omega_0) \to 1$ when the sample size goes to infinity.

Remark:

- This assumption requires the density ratio estimator $\widehat{\boldsymbol{\zeta}}$ to converge to $\boldsymbol{\zeta}_0$ in both ℓ_1 and ℓ_2 norms
- The ℓ_1 convergence is bounded by a constant, while ℓ_2 error vanishes asymptotically
- Can be satisfied by penalized estimation (e.g., Lasso) or ML methods (e.g., deep neural networks) when $N_{\mathcal{S}}, N_{\mathcal{T}}$ are sufficiently large
- Allows for model misspecification of $w(\mathbf{z})$

Structural Assumptions

- Sparsity: The bias δ_0, and parameters θ, β are sparse.
- Overlap: Source and Target distributions have sufficient overlap (bounded density ratio).
- Regularity: Sub-Gaussian tails and restricted eigenvalues.

Assumptions

Further Remarks about Assumption 1. The density ratio estimaton problem can be reformulated as a binary classification problem. Why? Define a binary domain indicator C, where $C = 0$ for Source samples and $C = 1$ for Target samples. By Bayes' theorem:

$$P(C = 1|\mathbf{z}) = \frac{p(\mathbf{z}|C = 1)P(C = 1)}{p(\mathbf{z})} = \frac{p_{\mathcal{T}}(\mathbf{z})\pi_{\mathcal{T}}}{p(\mathbf{z})} \tag{1}$$

$$P(C = 0|\mathbf{z}) = \frac{p(\mathbf{z}|C = 0)P(C = 0)}{p(\mathbf{z})} = \frac{p_{\mathcal{S}}(\mathbf{z})\pi_{\mathcal{S}}}{p(\mathbf{z})} \tag{2}$$

Taking the ratio of the posterior probabilities:

$$\frac{P(C = 1|\mathbf{z})}{P(C = 0|\mathbf{z})} = \frac{p_{\mathcal{T}}(\mathbf{z})\pi_{\mathcal{T}}}{p_{\mathcal{S}}(\mathbf{z})\pi_{\mathcal{S}}} = w(\mathbf{z}) \times \frac{\pi_{\mathcal{T}}}{\pi_{\mathcal{S}}} \tag{3}$$

Therefore, the density ratio $w(\mathbf{z})$ can be estimated via a classifier:

$$w(\mathbf{z}) = \frac{p_{\mathcal{T}}(\mathbf{z})}{p_{\mathcal{S}}(\mathbf{z})} = \frac{P(C = 1|\mathbf{z})}{P(C = 0|\mathbf{z})} \times \frac{N_{\mathcal{S}}}{N_{\mathcal{T}}} \tag{4}$$

(Note: $\pi_{\mathcal{S}}, \pi_{\mathcal{T}}$ are priors, can be estimated by $N_{\mathcal{S}}/N_{\mathcal{T}}$).

Theorem 1: Error Bound of Meta-Learning Estimator

Theorem

Suppose that Assumptions 1, and S1–S4 hold (in supplement materials). We have

$$\|\widehat{\boldsymbol{\theta}}^{(meta)} - \bar{\boldsymbol{\theta}}\|_1 = O_p\left(s_\theta\sqrt{\frac{\log d}{n}} + \|\boldsymbol{\delta}_0\|_1 + \sqrt{\frac{n}{\log d}}Err_\zeta\left(1 \vee \sqrt{\frac{n}{n_T}}Err_\zeta\right)\right)$$

$$\|\widehat{\boldsymbol{\theta}}^{(meta)} - \bar{\boldsymbol{\theta}}\|_2^2 = O_p\left(s_\theta\frac{\log d}{n} + \sqrt{\frac{\log d}{n_T}}\|\boldsymbol{\delta}_0\|_1 \wedge \|\boldsymbol{\delta}_0\|_1^2 + Err_\zeta\left(1 \vee \frac{n}{n_T}Err_\zeta\right)\right)$$

where $Err_\zeta = n_S^2 n^{-2}(\|\widehat{\boldsymbol{\zeta}} - \boldsymbol{\zeta}_0\|_2^2 + n_S^{-1}\log d\|\widehat{\boldsymbol{\zeta}} - \boldsymbol{\zeta}_0\|_1^2)$.

Remark:

- When $w(\mathbf{z})$ or $m(\mathbf{z})$ is correctly specified: $\boldsymbol{\delta}_0 = \mathbf{0}$, we get standard high-dimensional rates with combined sample size $n = n_{\mathcal{S}} + n_{\mathcal{T}}$
- **Double robustness**: source data fully used when either model is correct.

Theorem 2: Error Bound of the STRIFLE Estimator for Imputation Model

Theorem

Suppose that Assumptions 1, and S1–S4 hold. We have

$$\|\widehat{\boldsymbol{\theta}}^{(STRIFLE)} - \bar{\boldsymbol{\theta}}\|_2^2 = O_p\left[\left\{s_\theta \frac{\log d}{n} + \sqrt{\frac{\log d}{n_{\mathcal{T}}}}\|\boldsymbol{\delta}_0\|_1 \wedge \|\boldsymbol{\delta}_0\|_1^2 + Err_\zeta\left(1 \vee \frac{n}{n_{\mathcal{T}}} Err_\zeta\right)\right\} \wedge \left(\frac{\log d}{n_{\mathcal{T}}} s_\theta + n_{\mathcal{T}}^{-1/2}\right)\right]$$

Remark:

- **Triple robustness**: guaranteed to be no worse than target-only estimator when $n_{\mathcal{T}}^{1/2}/\log d \gtrsim s_\theta$
- When $s_\theta = o(n_{\mathcal{T}}^{1/2}/\log d)$: small price to pay for transferability detection

Theorem 3: Error Bound of Final STRIFLE Estimator for $\boldsymbol{\beta}_0$

Theorem

Under Assumptions 1, and S1–S5, let $\varpi = n_{\mathcal{T}}/(n_{\mathcal{T}} + N_{\mathcal{T}})$, we have

$$\|\widehat{\boldsymbol{\beta}}^{(STRIFLE)} - \boldsymbol{\beta}_0\|_2^2 = O_p\Bigg((1-\varpi)\Bigg[\Bigg\{s_\theta \frac{\log d}{n} + \sqrt{\frac{\log d}{n_{\mathcal{T}}}}\|\boldsymbol{\delta}_0\|_1 \wedge \|\boldsymbol{\delta}_0\|_1^2 + Err_\zeta\left(1 \vee \frac{n}{n_{\mathcal{T}}} Err_\zeta\right)\Bigg\} \wedge \left(\frac{\log d}{n_{\mathcal{T}}} s_\theta + n_{\mathcal{T}}^{-1/2}\right)\Bigg] + s_\beta \frac{\log p}{N_{\mathcal{T}} + n_{\mathcal{T}}}\Bigg)$$

Remark:

- Guaranteed no worse than target-only supervised estimator when $(1-\varpi)s_\theta n_{\mathcal{T}}^{-1} \log d \lesssim s_\beta n_{\mathcal{T}}^{-1} \log p$ and $(1-\varpi)n_{\mathcal{T}}^{1/2}/\log p \gtrsim s_\beta$
- Typically met in EHR applications: $\bar{\boldsymbol{\theta}}$ much sparser than $\boldsymbol{\beta}_0$
- When $s_\beta = o\{(1-\varpi)n_{\mathcal{T}}^{1/2}/\log p\}$: transferability detection error becomes dominant

Error Bound Comparisons with Different Methods

Table 1. Comparison of squared l_2 error bound of estimation of $\boldsymbol{\beta}_0$ among three methods in SS setting with negligible Err_ζ and $1-\varrho \asymp 1$.

Method	SAS	Meta	STRIFLE
$0 \le \|\boldsymbol{\delta}_0\|_1 \precsim \mathrm{cutoff}_1$	$\boldsymbol{\Delta}_{\boldsymbol{\beta}} + s_{\boldsymbol{\theta}} n_{\mathcal{T}}^{-1} \log d$	$\boldsymbol{\Delta}_{\boldsymbol{\beta}} + s_{\boldsymbol{\theta}} n^{-1} \log d$	$\boldsymbol{\Delta}_{\boldsymbol{\beta}} + s_{\boldsymbol{\theta}} n^{-1} \log d$
$\mathrm{cutoff}_1 \ll \|\boldsymbol{\delta}_0\|_1 \precsim \mathrm{cutoff}_2$	$\boldsymbol{\Delta}_{\boldsymbol{\beta}} + s_{\boldsymbol{\theta}} n_{\mathcal{T}}^{-1} \log d$	$\boldsymbol{\Delta}_{\boldsymbol{\beta}} + \sqrt{n_{\mathcal{T}}^{-1} \log d} \|\boldsymbol{\delta}_0\|_1$	$\boldsymbol{\Delta}_{\boldsymbol{\beta}} + \sqrt{n_{\mathcal{T}}^{-1} \log d} \|\boldsymbol{\delta}_0\|_1$
$\|\boldsymbol{\delta}_0\|_1 \gg \mathrm{cutoff}_2$	$\boldsymbol{\Delta}_{\boldsymbol{\beta}} + s_{\boldsymbol{\theta}} n_{\mathcal{T}}^{-1} \log d$	$\boldsymbol{\Delta}_{\boldsymbol{\beta}} + \sqrt{n_{\mathcal{T}}^{-1} \log d} \|\boldsymbol{\delta}_0\|_1$	$\boldsymbol{\Delta}_{\boldsymbol{\beta}} + s_{\boldsymbol{\theta}} n_{\mathcal{T}}^{-1} \log d + n_{\mathcal{T}}^{-1/2}$

NOTE: The common term $\boldsymbol{\Delta}_{\boldsymbol{\beta}} = s_{\boldsymbol{\beta}} (N_{\mathcal{T}} + n_{\mathcal{T}})^{-1} \log p$, $\mathrm{cutoff}_1 = s_{\boldsymbol{\theta}} n^{-1} \sqrt{n_{\mathcal{T}} \log d}$, and $\mathrm{cutoff}_2 = s_{\boldsymbol{\theta}} \sqrt{n_{\mathcal{T}}^{-1} \log d} \vee (\log d)^{-1/2}$.

Figure 1: Error bound comparisons

Cutoff 1: the dividing line of whether bias or variance dominate the error. Let

$$\sqrt{\frac{\log d}{n_{\mathcal{T}}}} \|\delta_0\|_1 = s_\theta \frac{\log d}{n}.$$

Cutoff 2: the dividing line of positive transfer and negative transfer. Let

$$\sqrt{\frac{\log d}{n_{\mathcal{T}}}} \|\delta_0\|_1 = s_\theta \frac{\log d}{n_{\mathcal{T}}}.$$

Content

1. Introduction

2. Background & Intuition

3. Derivation & Algorithms

4. Theoretical Results

5. Simulations & Real-Data Applications

Simulation Configurations

Sample Sizes:

- Target: $n_{\mathcal{T}} = 150$, $N_{\mathcal{T}} = 10{,}000$
- Source: $n_{\mathcal{S}} \in \{150, 1200\}$, $N_{\mathcal{S}} = 10{,}000$
- Dimensions: $p \in \{150, 1000\}$, $q = 30$

Covariate Generation:

1. Generate latent variables: $\boldsymbol{W}_i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ where $\boldsymbol{\Sigma}_{jk} = \nu^{|j-k|}$
   - $\nu = 0$: independent features
   - $\nu = 0.1$: weakly correlated features
2. Transform to zero-inflated discrete variables:

$$X_{ij} = [\log(1 + e^{W_{ij}})], \quad S_{i1}, S_{i2}, S_{i3}, S_{ij} (j = 4, \cdots, q)$$

   where $a(t) = [\log(1 + e^t)]$
3. Standardize $\mathbf{X}$ and $\boldsymbol{S}$ to mean 0 and variance 1

Purpose: Mimic EHR data characteristics (zero-inflated, discrete, correlated)

Simulation Configurations

Outcome Model Y (Binary):

- **(Mcor)** Correctly specified:

$$\text{logit}\{P(Y=1|\mathbf{Z})\} = 0.2 - X_1 + X_2 - X_3 + 2S_1 - 2S_2 + 2S_3$$

- **(Mmis)** Misspecified:

$$\text{logit}\{P(Y=1|\mathbf{Z})\} = -X_1 + X_2 + X_3^2 - S_2 + 2S_3 + \frac{2S_1}{1+\exp(-X_1 S_1^2)}$$

- **(M'mis)** Severely misspecified:

$$\text{logit}\{P(Y=1|\mathbf{Z})\} = -1 - 2X_1 + 2X_2^2 + 2X_3X_4 + 4S_1^2 - 4S_2^2 + \cdots$$

Population Membership Model R (Source = 1):

- **(Wcor)** Correctly specified: $\text{logit}\{P(R=0|\mathbf{Z})\} = X_1 - X_2 - X_3 + S_1$
- **(Wmis)** Misspecified: complex nonlinear interactions
- **(W'mis)** Severely misspecified: more complex interactions

Simulation Configurations

Five Configurations Testing Triple Robustness:

Config	Y Model	R Model	Purpose
(C1)	(Mcor) Correct	(Wcor) Correct	Both correct Full transfer expected
(C2)	(Mmis) Wrong	(Wcor) Correct	$\boldsymbol{\delta}_0 = \mathbf{0}$ Density ratio saves
(C3)	(Mcor) Correct	(Wmis) Wrong	$\boldsymbol{\delta}_0 = \mathbf{0}$ Imputation saves
(C4)	(Mmis) Wrong	(Wmis) Wrong	Mild misspec. Partial transfer expected
(C5)	(M'mis) Very Wrong	(W'mis) Very Wrong	Severe misspec. Negative transfer test

Each configuration combined with: $\nu \in \{0, 0.1\}$, $p \in \{150, 1000\}$, $n_{\mathcal{S}} \in \{150, 1200\}$
Total: 5 configs $\times$ 2 correlations $\times$ 2 dimensions $\times$ 2 source sizes $=$ 40 scenarios
Evaluation: 500 replications, metrics $=$ |Bias|, l_2 Error, AUC

Simulation Studies

	|Bias|	l_2 Err	AUC	|Bias|	l_2 Err	AUC	|Bias|	l_2 Err	AUC
$\nu = 0$	(C1) Both correct			(C2) $m(\mathbf{z})$ misspecified			(C3) $w(\mathbf{z})$ misspecified		
SUP	0.86	1.16	0.72	0.99	1.39	0.62	0.92	1.33	0.61
SAS	0.81	1.06	0.75	0.93	1.29	0.67	0.84	1.16	0.69
$\text{CS}_{n_{\mathcal{S}}=150}$	0.84	1.11	0.73	0.96	1.31	0.65	0.89	1.27	0.63
$\text{SUPTrans}_{n_{\mathcal{S}}=150}$	0.89	1.23	0.73	1.01	1.39	0.66	0.94	1.28	0.64
$\text{STRIFLE}_{n_{\mathcal{S}}=150}$	**0.78**	**0.95**	**0.76**	**0.89**	**1.18**	**0.70**	**0.81**	**1.10**	**0.70**
$\text{CS}_{n_{\mathcal{S}}=1200}$	0.77	0.84	0.77	0.84	0.93	0.73	0.85	1.08	0.68
$\text{SUPTrans}_{n_{\mathcal{S}}=1200}$	0.82	0.95	0.79	0.90	1.02	**0.76**	0.88	1.07	0.69
$\text{STRIFLE}_{n_{\mathcal{S}}=1200}$	**0.72**	**0.64**	**0.80**	**0.79**	**0.78**	0.76	**0.74**	**0.77**	**0.75**
$\nu = 0.1$	(C1) Both correct			(C2) $m(\mathbf{z})$ misspecified			(C3) $w(\mathbf{z})$ misspecified		
SUP	0.87	1.20	0.71	1.01	1.40	0.63	0.95	1.38	0.60
SAS	0.83	1.11	0.73	0.95	1.29	0.69	0.86	1.23	0.69
$\text{CS}_{n_{\mathcal{S}}=150}$	0.85	1.14	0.72	0.97	1.30	0.67	0.91	1.32	0.63
$\text{SUPTrans}_{n_{\mathcal{S}}=150}$	0.91	1.28	0.71	1.02	1.40	0.67	0.96	1.34	0.63
$\text{STRIFLE}_{n_{\mathcal{S}}=150}$	**0.80**	**1.00**	**0.75**	**0.90**	**1.18**	**0.71**	**0.84**	**1.16**	**0.70**
$\text{CS}_{n_{\mathcal{S}}=1200}$	0.77	0.84	0.77	0.85	0.92	0.75	0.86	1.09	0.69
$\text{SUPTrans}_{n_{\mathcal{S}}=1200}$	0.83	0.96	0.79	0.91	1.02	**0.77**	0.90	1.11	0.70
$\text{STRIFLE}_{n_{\mathcal{S}}=1200}$	**0.73**	**0.67**	**0.80**	**0.80**	**0.80**	0.77	**0.75**	**0.80**	**0.75**

NOTE: Bold values highlight the best-performing methods based on the corresponding metrics.

Figure 2: Numerical Performance

Simulation Studies

	|Bias|	l_2 Err	AUC	|Bias|	l_2 Err	AUC
$\nu = 0$	(C4) $(M_{\text{mis}}) + (W_{\text{mis}})$			(C5) $(M'_{\text{mis}}) + (W'_{\text{mis}})$		
SUP	1.03	1.51	0.61	0.94	1.32	0.73
SAS	0.90	1.24	0.73	**0.86**	**1.08**	**0.78**
$\text{CS}_{n_{\mathcal{S}}=150}$	1.00	1.42	0.65	1.09	1.91	0.63
$\text{SUPTrans}_{n_{\mathcal{S}}=150}$	1.04	1.53	0.62	0.96	1.35	0.73
$\text{Meta}_{n_{\mathcal{S}}=150}$	**0.86**	**1.09**	**0.75**	0.95	1.39	0.73
$\text{STRIFLE}_{n_{\mathcal{S}}=150}$	0.87	1.11	0.74	0.87	1.12	0.77
$\text{CS}_{n_{\mathcal{S}}=1200}$	0.90	1.09	0.74	1.18	2.68	0.60
$\text{SUPTrans}_{n_{\mathcal{S}}=1200}$	0.96	1.26	0.68	0.94	1.32	0.76
$\text{Meta}_{n_{\mathcal{S}}=1200}$	**0.78**	**0.75**	**0.79**	0.97	1.48	0.69
$\text{STRIFLE}_{n_{\mathcal{S}}=1200}$	0.78	0.76	0.79	0.87	1.13	0.77

NOTE: Negative transfer does not show up in (C4) but does exist in (C5). Bold values highlight the best-performing methods based on the corresponding metrics.

Figure 3: Numerical Performance

Real-Data Experiments

- **Target Population:** African Americans (AA) - small sample size
- **Source Population:** European ancestry - large sample size

Variables:

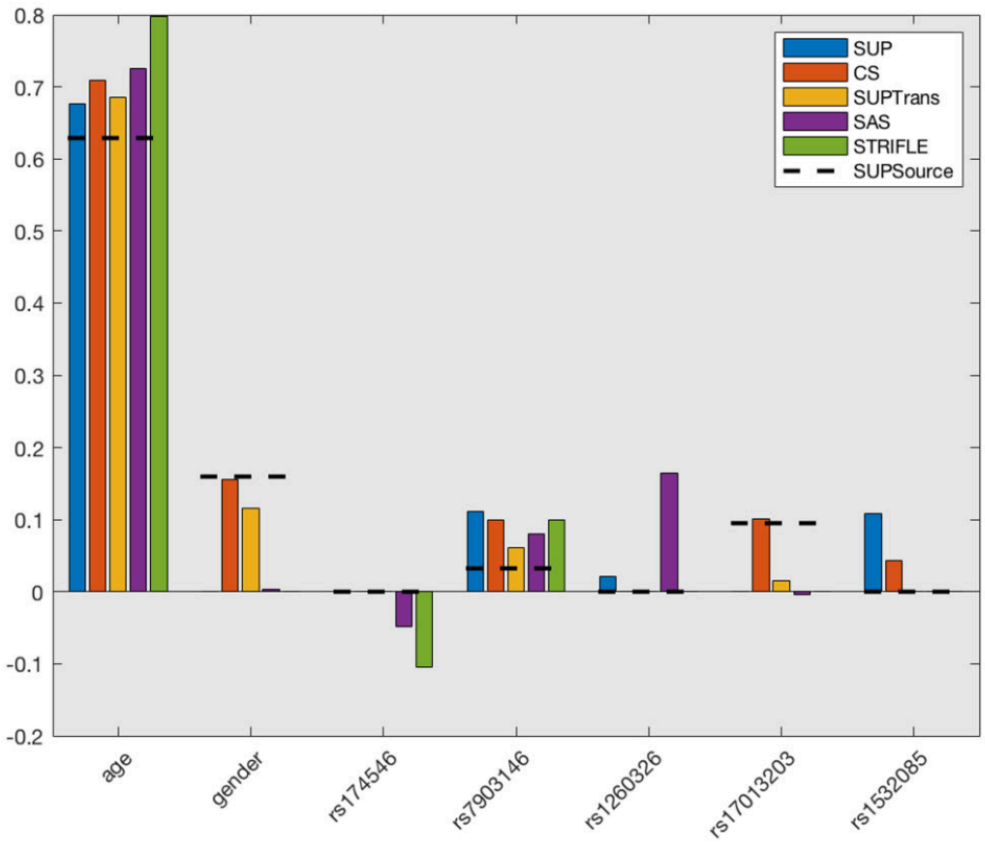
- Y **(Outcome):** T2D Status (Type 2 Diabetes). Gold standard confirmed via manual chart review. High cost, thus limited labeled data.
- **X (Predictors):** SNPs + Age + Gender. Contains 248 diabetes-related genetic variants (SNPs) plus age and gender. Input features for the risk prediction model.
- S **(Surrogates):** EHR Features (Labs & Meds). Contains 76 features, mainly lab test counts (e.g., HbA1c test frequency) and medication prescriptions (e.g., Metformin). Note: ICD diagnostic codes were intentionally excluded due to potential systematic coding bias across populations.

1. **Performance Improvement:** STRIFLE achieved AUC of **0.85**, significantly higher than Target-only SAS (0.76) and direct transfer SUPTrans (0.79). MSE (Mean Squared Error) was also lowest.
2. **Population Heterogeneity:**
   - **Gender:** Strong predictor in Source (European) population but weak effect in Target (AA) population. Direct transfer (SUPSource) incorrectly estimates gender's impact, while STRIFLE correctly adjusts for this.
   - **Key SNPs:** STRIFLE successfully identified important SNPs for AA population, such as **rs7903146** (known T2D susceptibility locus) and **rs174546** (related to fatty acid desaturation and enzyme activity).

Real-Data Experiments

	Source estimate	Target estimate				
	SUPSource	SUP	CS	SUPTrans	SAS	STRIFLE
AUC	0.77	0.66	0.77	0.76	0.79	**0.85**
MSE	0.21	0.21	0.21	0.20	0.19	**0.17**

NOTE: Bold values highlight the best-performing methods based on the corresponding metrics.



Thanks!